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Structural, magnetic and electromagnetic wave absorption properties of La-BaM/PANI nanocomposites

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Introduction

- Hexaferites are classified into six groups: M-, Z-, Y-, W-, X- and U-types.
- Magnetic properties of hexaferrites can be modified by substitution A and B sites by other elements.
- Hexaferrites are concerned to be the best choice for industrial purpose because of their high resistivity, high saturation as well as excellent electromagnetic properties.
- Composites of M-type hexaferrites/conductive polymers can be used to enhance the microwave absorption in the minimum reflection loss (RL) as well as the effective bandwidth absorption.
- To understand the role of doping and composite with conductive polymer on the structural, magnetic and microwave absorption properties of M-type hexaferites, composites of La-doped $BaFe_{12}O_{19}$ and polyaniline (PANI) were prepared.

Experiment

- $Ba_{1-x}La_xFe_{12}O_{19}$ (La-BaM, x = 0 – 0.5) samples were prepared by co-precipitation combined with heat treatment.
- Composites were prepared by mixing La-BaM with PANI and epoxy following by drying at 170 °C for 0.5 h.
- Morphology was checked by FE-SEM.
- Crystal structure was checked by using XRD.
- Magnetic measurements were carried out using a VSM.
- Microwave absorption measurements were performed by Vector Network Analyzer (VNA).

Results & Discussion

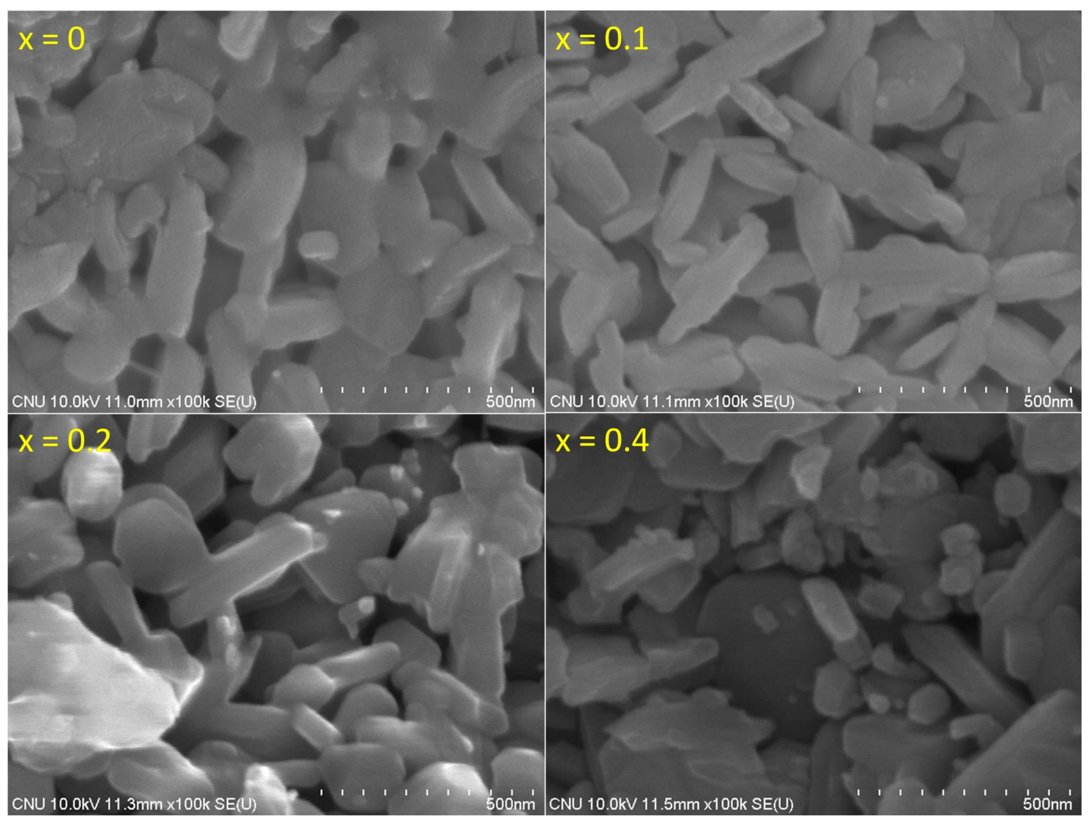

Fig. 1. FE-SEM images of some typical samples.

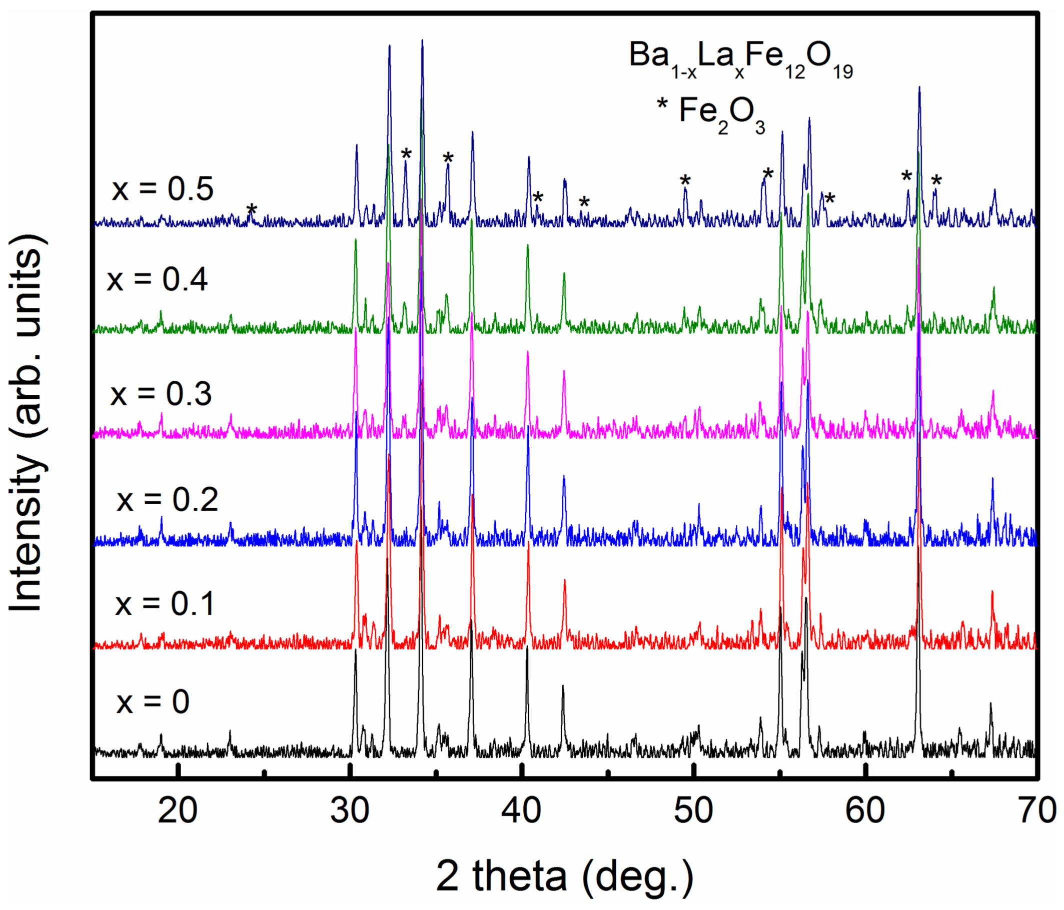

Fig. 2. XRD patterns of $Ba_{1-x}La_xFe_{12}O_{19}$ samples.

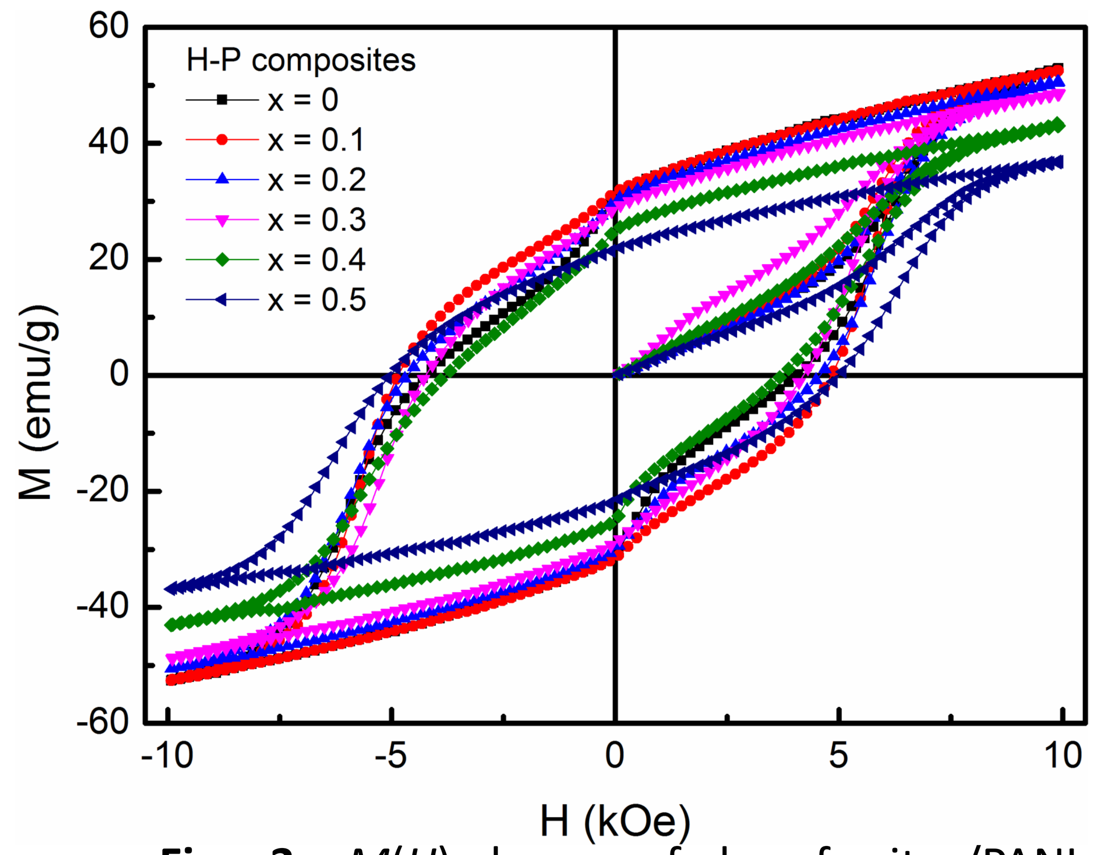

Fig. 3. $M(H)$ loops of hexaferrites/PANI composites recorded at room temperature.

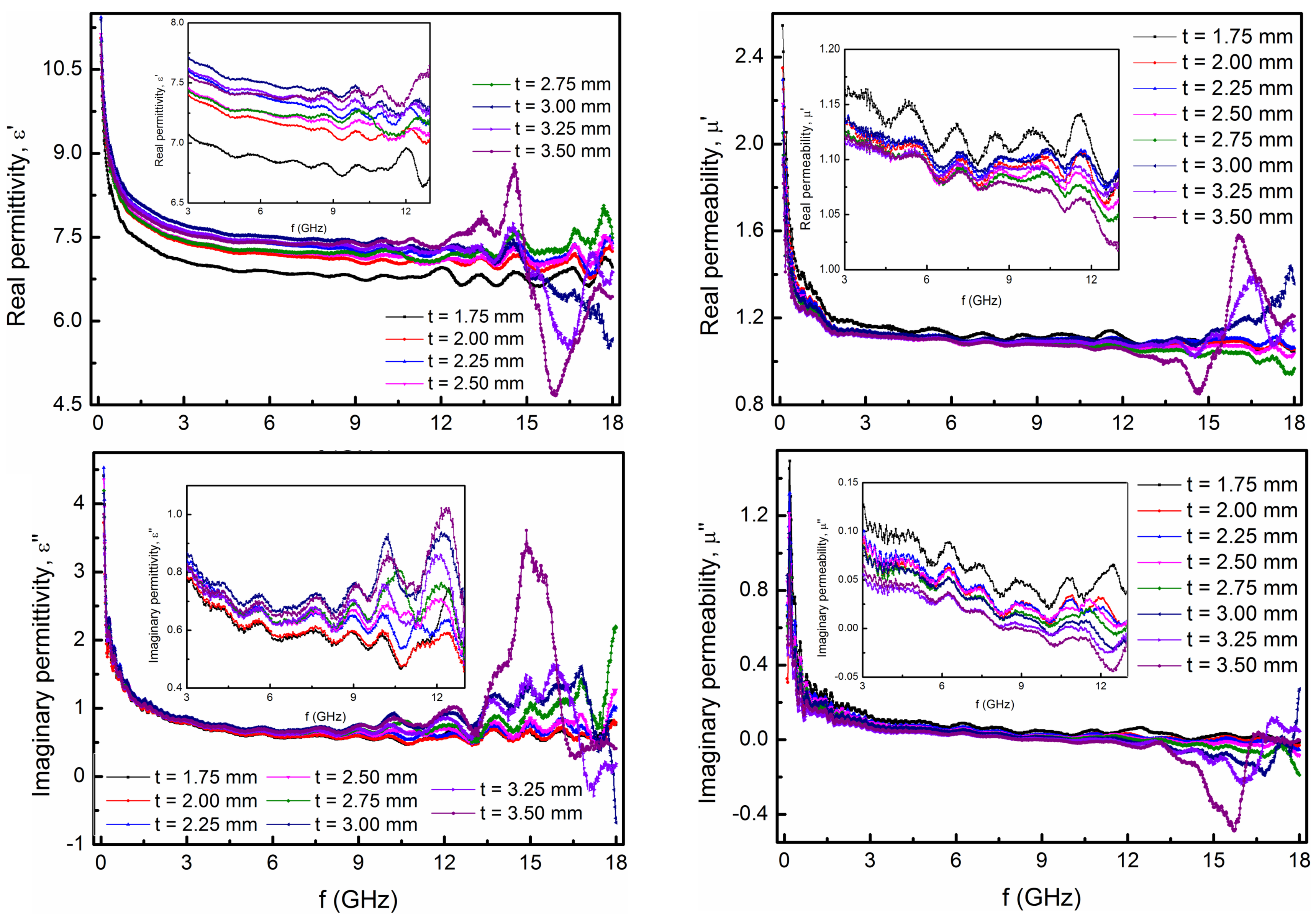

Fig. 4. Complex permittivity (*left*) and complex permeability (*right*) of La-BaM-0/PANI composite. Insets: Enlarged views in the range of 3 – 13 GHz.

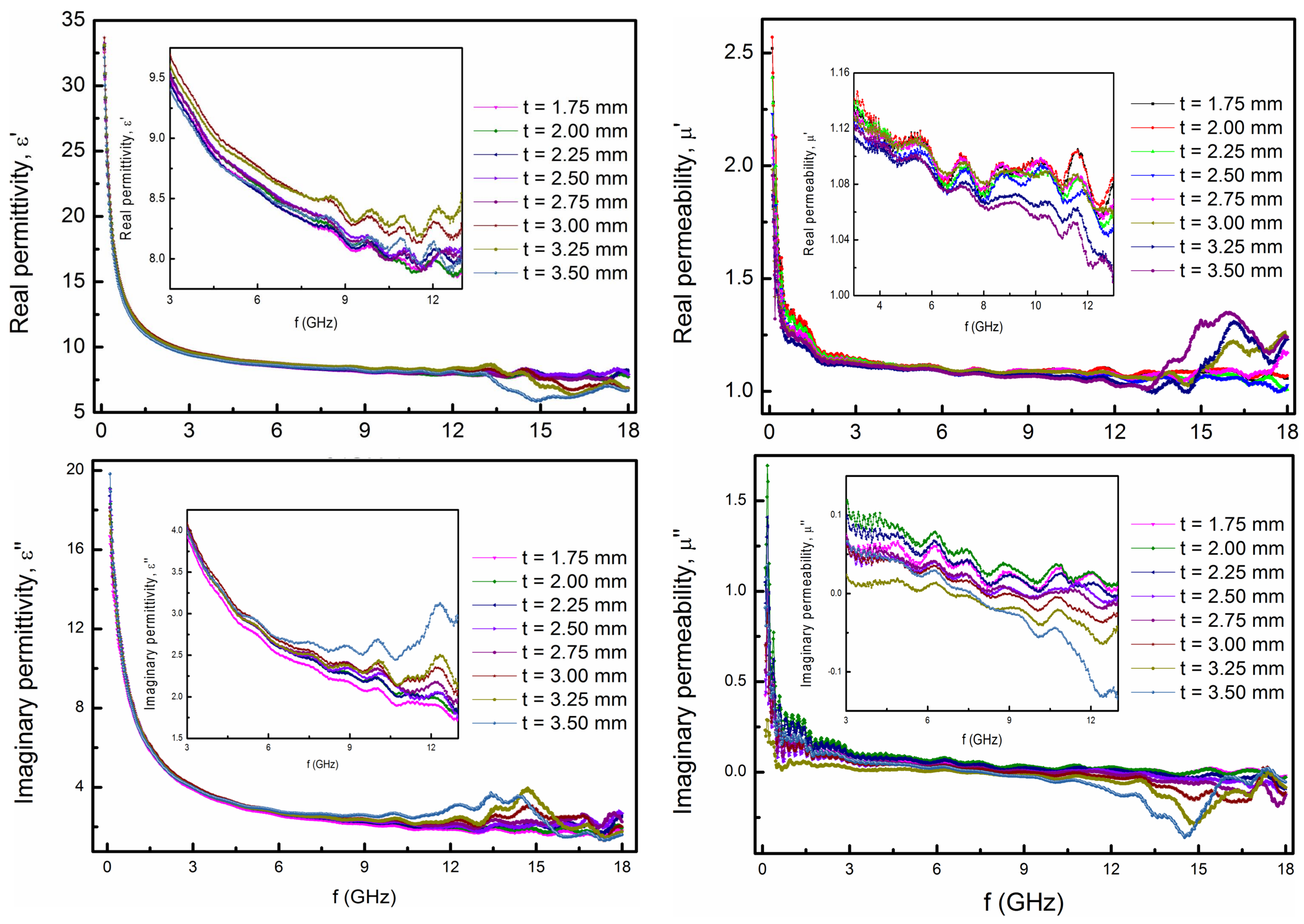

Fig. 5. Complex permittivity (*left*) and complex permeability (*right*) of La-BaM-0.1/PANI composite. Insets: Enlarged views in the range of 3 – 13 GHz.

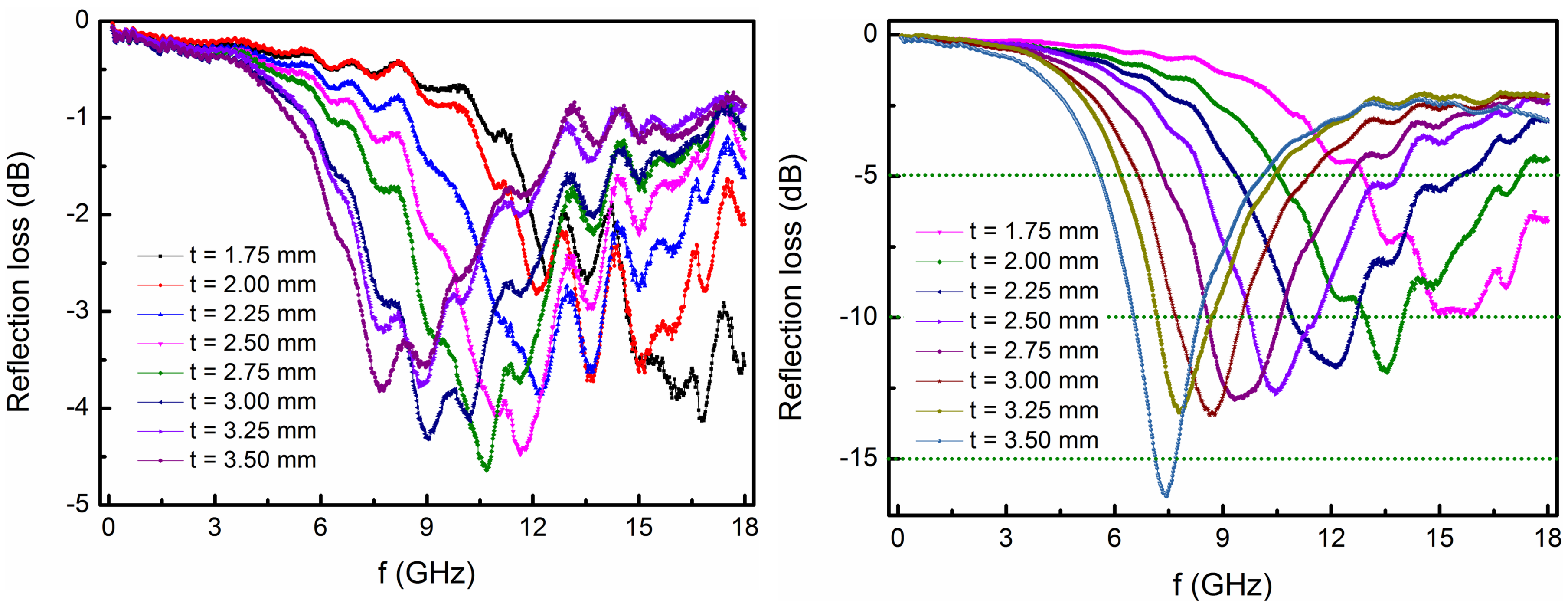

Fig. 6. Reflection loss vs thickness t of La-BaM/PANI composites: x = 0 (*left*) and x = 0.1 (*right*).

- Both real and imaginary parts of permittivity of composites had been change by doping: much changes for x = 0.1; less changes for x = 0.3 and 0.5.
- While, both parts of complex permeability were almost unchanged

Summary

- Morphology was changed by La doping.
- XRD patterns had showed a second phase of Fe_2O_3 with high-doping levels → decreasing M_s.
- By doping, complex permittivity of composites was changed → changing of reflectivity.
- The best microwave absorption can be found in composites x = 0.1.